DP-900

Microsoft Azure Data Fundamentals

Complete Study Notes · 2025

■ Exam At a Glance

Item	Detail
Exam Code	DP-900
Full Name	Microsoft Azure Data Fundamentals
Passing Score	700 / 1000
Duration	~45 minutes
Questions	~40–60 (scenario-based)
Cost	USD 165 (varies by region)

■ Exam Weightage & Notes Coverage

Section	Weight	Days	In This PDF
1. Core Data Concepts	25–30%	Day 1–2	Section 1
2. Relational Data on Azure	20–25%	Day 3–4	Section 2
3. Non-Relational Data on Azure	15–20%	Day 5–6	Section 3
4. Analytics Workload on Azure	25–30%	Day 7–8	Section 4
5. Cosmos DB Deep Dive	Part of Sec 3	Day 5	Section 5 NEW
6. Synapse vs Databricks vs Fabric	Part of Sec 4	Day 7	Section 6 NEW
7. Power BI Deep Dive	Part of Sec 4	Day 8	Section 7 NEW
8. ADF, Security, ACID, ADLS Gen2	Across all	Day 6-7	Section 8 NEW
9. Practice Questions (10 Q+A)	All sections	Day 9	Section 9 NEW
10. Revision + Real Practice Tests	—	Day 10	External

■ *Sections 1 and 4 carry the most weight — prioritise them!*

SECTION 1 — Core Data Concepts (25–30%)

■ Types of Data

Type	Description	Example
Structured	Fixed schema, rows & columns	SQL tables, CSV
Semi-structured	Flexible schema, self-describing	JSON, XML, Parquet
Unstructured	No schema at all	Images, videos, PDFs

■ Common File Formats

Format	Type	Best For
CSV	Structured	Simple tabular data, human-readable
JSON	Semi-structured	APIs, flexible key-value data
Parquet	Semi-structured (columnar)	Analytics, big data — compressed & fast
Avro	Semi-structured (row-based)	Streaming, schema evolution
ORC	Structured (columnar)	Hadoop/Hive workloads

■ Database Types

DB Type	Description	Azure Example
Relational	Tables with rows/columns, SQL	Azure SQL Database
NoSQL / Document	Flexible JSON documents	Cosmos DB (NoSQL API)
Key-Value	Simple key → value pairs	Azure Table Storage
Graph	Entities + relationships	Cosmos DB (Gremlin API)
Time-series	Data ordered by timestamp	Azure Data Explorer
Column-family	Wide-column store	Cosmos DB (Cassandra API)

■ OLTP vs OLAP — Very Important!

	OLTP (Transactional)	OLAP (Analytical)
Purpose	Day-to-day operations	Historical analysis & reporting
Query type	Many small, fast read/write	Few large, complex queries
Data volume	Current data	Large historical datasets
Azure service	Azure SQL Database	Azure Synapse Analytics
Example	Online order processing	Monthly sales dashboard

■ Data Roles — Frequently Tested!

Role	Key Responsibilities
Database Administrator	Manage, secure, backup & restore databases; handle permissions
Data Engineer	Build data pipelines, ingest & transform data, maintain infrastructure
Data Analyst	Query data, create reports & dashboards, derive business insights

■ *Memory trick: DBA = security & uptime | Engineer = pipelines & movement | Analyst = insights & reports*

SECTION 2 — Relational Data on Azure (20–25%)

■ Key Relational Concepts

- **Tables** store data in rows (records) and columns (attributes).
- **Primary Key (PK)** — uniquely identifies each row in a table.
- **Foreign Key (FK)** — links one table to another, enforces referential integrity.
- **Index** — speeds up query lookups (like a book index).
- **View** — a virtual table defined by a SELECT query.
- **Stored Procedure** — saved, reusable SQL logic block.
- **Normalization** — organising data to reduce redundancy.

■ Normalization Forms

Form	Rule
1NF	No repeating groups; each cell has one value
2NF	1NF + no partial dependencies on composite keys
3NF	2NF + no transitive dependencies (non-key on non-key)

■ SQL Statement Categories

Category	Statements	Purpose
DML — Data Manipulation	SELECT, INSERT, UPDATE, DELETE	Read/write data
DDL — Data Definition	CREATE, ALTER, DROP, TRUNCATE	Define structure
DCL — Data Control	GRANT, REVOKE	Manage permissions
TCL — Transaction Control	COMMIT, ROLLBACK	Control transactions

■ Azure Relational Services — Must Know!

Service	Type	When to Use
Azure SQL Database	PaaS — Fully managed	New cloud-native apps, serverless option
Azure SQL Managed Instance	PaaS — Near 100% SQL Server compat.	Lift-and-shift from on-prem SQL Server
SQL Server on Azure VM	IaaS — Full control	Need full OS access or legacy features
Azure DB for MySQL	PaaS — Managed MySQL	Open-source MySQL applications
Azure DB for PostgreSQL	PaaS — Managed PostgreSQL	Open-source Postgres workloads
Azure DB for MariaDB	PaaS — Managed MariaDB	MariaDB community workloads

■ *PaaS = Microsoft handles patches, backups, HA. IaaS = YOU manage the OS and everything.*

SECTION 3 — Non-Relational Data on Azure (15–20%)

■ Azure Storage Services

Service	Data Type	Use Case
Azure Blob Storage	Unstructured (files, images, video)	Backups, media files, data lake storage
Azure File Storage	Files (SMB/NFS protocol)	Shared cloud file drive for VMs/apps
Azure Table Storage	Semi-structured (key-value)	Simple, cheap NoSQL key-value store
Azure Queue Storage	Messages	Async messaging between app components

■ Blob Storage Access Tiers

Tier	Access Frequency	Cost	Retrieval Speed
Hot	Frequent	Higher storage cost	Immediate
Cool	Infrequent (30+ days)	Lower storage cost	Immediate
Archive	Rare (180+ days)	Lowest storage cost	Hours (rehydration needed)

■ Azure Cosmos DB

A globally distributed, multi-model NoSQL database service with single-digit millisecond latency, automatic scaling, and 99.999% availability SLA.

Cosmos DB APIs — Frequently Tested!

API	Data Model	Best For
NoSQL (Core)	JSON documents	Default; new cloud-native NoSQL apps
MongoDB	JSON documents	Existing MongoDB applications
Cassandra	Wide-column	Existing Cassandra workloads
Gremlin	Graph (nodes + edges)	Social networks, fraud detection
Table	Key-value	Upgrading from Azure Table Storage
PostgreSQL	Relational + distributed	Distributed PostgreSQL workloads

■ *Think of Cosmos DB as one service that speaks many 'languages' (APIs). Pick the API based on your existing app or data model.*

SECTION 4 — Analytics Workload on Azure (25–30%)

■ Modern Analytics Architecture

Stage	Description	Azure Service
Ingest	Collect raw data from sources	Azure Data Factory, Event Hubs
Store	Hold raw & processed data	ADLS Gen2, Azure Blob Storage
Process	Transform & clean data	Azure Synapse, Databricks, ADF
Serve	Make analytics-ready data available	Azure Synapse SQL, Power BI Datasets
Visualize	Report and dashboards	Power BI

■ Key Azure Analytics Services

Service	What it Does	Key Fact
Azure Data Factory (ADF)	Orchestrate & move data (ETL/ELT pipelines)	Code-free data integration
Azure Synapse Analytics	All-in-one: SQL + Spark + pipelines + Power BI	Unified analytics platform
Azure Databricks	Apache Spark-based analytics & ML	Best for ML/AI workloads
Microsoft Fabric	End-to-end unified analytics (newest)	Likely on exam — covers whole data lifecycle
Azure HDInsight	Managed Hadoop/Spark/Hive/HBase clusters	Open-source big data
Azure Stream Analytics	Real-time SQL-based stream processing	Queries on live data streams
Azure Event Hubs	High-throughput event ingestion service	Ingest millions of events/sec
Azure Data Explorer	Fast analytics on logs & time-series data	Telemetry & IoT analytics

■ Batch vs Streaming

	Batch Processing	Stream Processing
When processed	On a schedule (hourly, daily)	Continuously, in real-time
Latency	High (minutes to hours)	Low (milliseconds to seconds)
Data volume	Large historical chunks	Small, continuous events
Use case	Monthly sales report, ETL jobs	Fraud detection, live dashboards
Azure services	ADF, Synapse Spark	Event Hubs + Stream Analytics

■ Data Stores for Analytics

Store Type	Description	Azure Example
Data Lake	Raw storage for all data types at scale	Azure Data Lake Storage Gen2

Store Type	Description	Azure Example
Data Warehouse	Structured, query-optimised historical store	Azure Synapse Analytics
Data Lakehouse	Combines lake flexibility + warehouse performance	Microsoft Fabric / Databricks
OLAP Cube	Pre-aggregated multidimensional data	Analysis Services

■ Power BI — Definitely on the Exam!

Component	Description
Power BI Desktop	Windows app to build and design reports
Power BI Service	Cloud portal to publish, share, and collaborate
Power BI Mobile	View dashboards on iOS/Android
Dataset	The data source connected to Power BI
Report	Visualizations built on top of a dataset
Dashboard	Pinned tiles/visuals from one or more reports
Workspace	Collaborative area to share content with teams

Common Chart Types in Power BI

Chart	Best For
Bar / Column Chart	Comparing categories
Line Chart	Trends over time
Pie / Donut Chart	Part-to-whole proportions
Map / Filled Map	Geographic distribution
Card / KPI	Single headline metric
Scatter Plot	Correlation between two measures
Waterfall Chart	Running total with increases/decreases

■ *Power BI workflow: Connect data → Model (relationships) → Build report → Publish to Service → Share dashboard*

SECTION 5 — Cosmos DB Deep Dive (Frequently Tested)

■ Cosmos DB Consistency Levels — Very Exam-Heavy!

Cosmos DB offers 5 consistency levels, balancing between performance and data accuracy. Listed from strongest to weakest:

Level	Guarantee	Performance	Use Case
Strong	Always reads latest write — 100% accurate	Slowest	Financial transactions, inventory
Bounded Staleness	Reads lag behind writes by a defined amount	Moderate	Global apps needing near-consistency
Session (Default)	Consistent within a single client session	Good	Shopping carts, user profiles
Consistent Prefix	Reads never see out-of-order writes	Better	Social media feeds, logs
Eventual	No ordering guarantee — eventually consistent	Fastest	Likes/views counters, leaderboards

■ *Memory trick — Strong → Bounded → Session → Prefix → Eventual. Default is SESSION. Strongest = slowest. Weakest = fastest.*

■ Cosmos DB Key Properties

Property	Detail
Latency SLA	< 10ms reads, < 15ms writes at 99th percentile
Availability SLA	99.999% for multi-region write accounts
Scaling	Automatic, serverless, or manual throughput (RU/s)
Throughput unit	Request Units per second (RU/s) — measures cost of operations
Partition Key	Determines how data is distributed across partitions — choose wisely!
Global Distribution	Replicate data to any Azure region with one click

■ Cosmos DB APIs — Full Comparison

API	Protocol	Data Model	Migrate From	Best For
NoSQL (Core)	HTTP/REST	JSON documents	New apps	Default — flexible JSON
MongoDB	MongoDB wire	BSON documents	MongoDB	Existing Mongo apps
Cassandra	CQL (Cassandra)	Wide columns	Cassandra	Column-family workloads
Gremlin	WebSocket/Gremlin	Graph	Graph DBs	Relationships & networks

API	Protocol	Data Model	Migrate From	Best For
Table	OData/REST	Key-value	Azure Table Storage	Simple key-value upgrade
PostgreSQL	PostgreSQL wire	Relational	PostgreSQL	Distributed relational

SECTION 6 — Synapse vs Databricks vs Fabric (Comparison Questions)

■ Side-by-Side Comparison

Feature	Azure Synapse Analytics	Azure Databricks	Microsoft Fabric
Type	Unified analytics platform	Apache Spark platform	End-to-end SaaS analytics
Primary use	SQL DW + Spark + Pipelines	Big data, ML, AI	Full data lifecycle (newest)
SQL support	Dedicated + Serverless SQL	Spark SQL only	Yes (via Warehouse & Lakehouse)
Spark support	Yes (Synapse Spark)	Yes (optimised Apache Spark)	Yes (via Spark notebooks)
Pipelines/ETL	Synapse Pipelines (like ADF)	Delta Live Tables	Data Factory in Fabric
ML/AI focus	Basic ML integration	Strong (MLflow built-in)	Moderate
Storage format	Parquet, Delta	Delta Lake (native)	OneLake (Delta/Parquet)
Power BI integration	Direct integration	Via connector	Native (built-in)
Best for exam Q	'All-in-one SQL + Spark'	'Spark-based ML/AI'	'Unified modern platform'

■ When to Choose Which — Exam Scenarios

Scenario	Choose
Need SQL data warehouse + Spark in one place	Azure Synapse Analytics
Running Apache Spark jobs, training ML models	Azure Databricks
Want a single unified platform covering ingestion to BI	Microsoft Fabric
Have existing Azure Data Factory pipelines	Azure Synapse (has built-in pipelines like ADF)
Need MLflow, Delta Lake natively optimised	Azure Databricks
Small team, want everything managed end-to-end	Microsoft Fabric

■ Microsoft Fabric — Key Components

Microsoft Fabric is the newest unified analytics platform. It brings together all data tools under one roof with OneLake as the single storage layer.

Fabric Component	What it Does
OneLake	Single unified data lake storage for entire organisation (like OneDrive for data)
Data Factory	Data integration and pipeline orchestration

Fabric Component	What it Does
Synapse Data Engineering	Spark-based data engineering with notebooks
Synapse Data Warehouse	SQL-based cloud data warehouse
Synapse Data Science	ML model building and experimentation
Real-Time Intelligence	Stream analytics and real-time dashboards
Power BI	Business intelligence and visualisation
Data Activator	Trigger actions automatically based on data conditions

■ *Fabric exam tip: OneLake = central storage. Everything else in Fabric reads/writes to OneLake. Think of it as 'one copy of data, many tools'.*

SECTION 7 — Power BI Deep Dive (High Exam Weight)

■ DirectQuery vs Import Mode — Frequently Tested!

	Import Mode	DirectQuery Mode	Live Connection
How it works	Data copied into Power BI model	Queries sent to source in real-time	Live connection to Analysis Services
Data freshness	Scheduled refresh (up to 8x/day)	Always real-time	Always real-time
Performance	Fastest (data in-memory)	Depends on source speed	Depends on source
Data size limit	~1GB compressed (Pro)	No limit	No limit
Transformations	Full Power Query support	Limited transformations	None — model is external
Best for	Most reports, smaller datasets	Large/real-time data	Existing Analysis Services models

■ *Default is Import Mode. Use DirectQuery when data must always be current or dataset is too large to import.*

■ Power BI Data Refresh

Refresh Type	Description	Max Frequency
Scheduled Refresh	Automatic refresh on a set schedule	8x per day (Pro), 48x (Premium)
On-demand Refresh	Manual refresh triggered by user	Any time
Incremental Refresh	Only refreshes new/changed data	Per policy settings
Real-time (DirectQuery)	No refresh needed — always live	Continuous

■ Power BI Licensing

License	What You Can Do
Power BI Free	Build reports in Desktop, publish to My Workspace only
Power BI Pro	Share, collaborate, publish to shared workspaces (USD 10/user/month)
Power BI Premium Per User	Premium features per user — larger models, more refreshes
Power BI Premium (Capacity)	Organisation-wide, viewers don't need Pro license

■ Power BI Data Model Concepts

Concept	Description
Star Schema	Central fact table surrounded by dimension tables — recommended for Power BI
Fact Table	Contains measurable events/transactions (sales, orders) — has foreign keys

Concept	Description
Dimension Table	Contains descriptive attributes (date, product, customer) — has primary key
Relationship	Link between tables — usually one-to-many from dimension to fact
Measure	Calculated aggregation using DAX (e.g. Total Sales = SUM(Sales[Amount]))
Calculated Column	Row-by-row calculation stored in the table
DAX	Data Analysis Expressions — formula language used in Power BI

■ *Star schema = best practice for Power BI. Fact tables are wide (many rows), dimension tables are narrow (fewer rows, descriptive).*

SECTION 8 — Additional Topics (Exam Coverage)

■ Data Ingestion & Processing Concepts

Concept	Description
ETL	Extract → Transform → Load. Transform before loading into warehouse (traditional)
ELT	Extract → Load → Transform. Load raw first, transform inside warehouse (modern)
Batch Ingestion	Collect data in chunks on a schedule (ADF, Synapse Pipelines)
Stream Ingestion	Ingest data continuously as it arrives (Event Hubs, IoT Hub)
Data Pipeline	Series of steps to move and transform data from source to destination
Partitioning	Splitting large datasets into smaller chunks for parallel processing

■ Azure Data Lake Storage Gen2 (ADLS Gen2)

Feature	Detail
What it is	Azure Blob Storage + hierarchical namespace = data lake capability
Hierarchical namespace	Folders and files (like a file system) instead of flat blob containers
Best for	Big data analytics — works natively with Synapse, Databricks, HDInsight
Access control	POSIX-compliant ACLs at file/folder level
Format support	Parquet, CSV, JSON, Delta, ORC — any file format
Key advantage over Blob	Much faster for analytics due to directory operations

■ Azure Data Factory (ADF) — Key Concepts

ADF Concept	Description
Pipeline	Logical grouping of activities that perform a task
Activity	A single step in a pipeline (Copy, Transform, Execute, etc.)
Dataset	Named reference to the data you want to use (input or output)
Linked Service	Connection string to a data source (like a database connector)
Integration Runtime	The compute infrastructure ADF uses to run activities
Trigger	Defines when a pipeline runs (schedule, event, tumbling window)
Copy Activity	Most common — moves data from source to sink

■ *ADF analogy: Pipeline = recipe, Activity = cooking step, Linked Service = ingredient source, Dataset = specific ingredient.*

■ Real-Time Analytics Architecture

Stage	Service	Role
Event generation	IoT devices, apps, clickstreams	Produce real-time events
Event ingestion	Azure Event Hubs / IoT Hub	Capture & buffer high-volume streams
Stream processing	Azure Stream Analytics	Query & transform live data with SQL
Output / Storage	Cosmos DB, SQL DB, Blob, Power BI	Store results or update dashboards
Visualisation	Power BI Streaming Dataset	Real-time live dashboard tiles

■ Security & Compliance Concepts

Concept	Description
Authentication	Verifying WHO you are (Azure AD, username/password)
Authorisation	Verifying WHAT you can do (RBAC, permissions)
Encryption at rest	Data encrypted when stored on disk (default in Azure)
Encryption in transit	Data encrypted while moving over network (TLS/SSL)
Azure RBAC	Role-Based Access Control — assign roles to users/groups
Transparent Data Encryption (TDE)	Automatic encryption for Azure SQL databases
Row-Level Security (RLS)	Restrict which rows a user can see in Power BI or SQL
Private Endpoint	Connect to Azure services over private network, not public internet

■ Transactional Properties — ACID

Property	Meaning	Example
Atomicity	All operations succeed or all fail — no partial updates	Bank transfer: debit + credit both happen or neither
Consistency	Data always moves from one valid state to another	Balance can never go below zero if rule exists
Isolation	Concurrent transactions don't interfere with each other	Two users booking last seat — only one succeeds
Durability	Committed transactions survive system failures	Power outage after commit — data is not lost

■ *ACID = property of OLTP / relational databases. Not guaranteed in NoSQL / eventual consistency systems.*

■ HDInsight — Open Source Big Data

Cluster Type	Technology	Use Case
Hadoop	MapReduce + HDFS	Batch processing of large datasets
Spark	In-memory processing	Fast batch + streaming analytics

Cluster Type	Technology	Use Case
HBase	Column-family NoSQL	Low-latency access to large tables
Kafka	Distributed streaming	High-throughput event streaming
Hive	SQL over Hadoop	Data warehousing on Hadoop
Storm	Real-time stream processing	Event-driven stream processing

■ *HDInsight exam tip: Used when you need managed open-source tools (Hadoop/Spark/Kafka) WITHOUT vendor lock-in. Databricks is preferred for Spark-only workloads.*

SECTION 9 — Practice Questions with Answers

1. You need to store product images and videos for an e-commerce app. Which Azure service should you use?

A) Azure Table Storage
B) Azure Blob Storage
C) Azure SQL Database
D) Azure File Storage

Answer: B) Azure Blob Storage	*Blob Storage is designed for unstructured data like images, videos, and documents.*

2. Your company has an on-premises SQL Server with features like SQL Agent and linked servers. You want to migrate to Azure with minimal changes. Which service is best?

A) Azure SQL Database
B) SQL Server on Azure VM
C) Azure SQL Managed Instance
D) Azure Cosmos DB

Answer: C) Azure SQL Managed Instance	*SQL MI provides near 100% compatibility with on-prem SQL Server, including features SQL Database doesn't support.*

3. A data engineer needs to build a pipeline that moves data from an FTP server to Azure Blob Storage on a daily schedule. Which service should they use?

A) Azure Stream Analytics
B) Azure Data Factory
C) Azure Databricks
D) Azure Synapse Spark

Answer: B) Azure Data Factory	*ADF is the orchestration tool for scheduled data movement and ETL pipelines.*

4. Which consistency level in Cosmos DB provides the strongest guarantee but the lowest performance?

A) Session
B) Eventual
C) Strong
D) Bounded Staleness

Answer: C) Strong	*Strong consistency always returns the most recent committed write — guaranteed, but slowest.*

5. You are building a social network app that needs to find connections between users (friends of friends). Which Cosmos DB API is most appropriate?

A) NoSQL API
B) Table API
C) Gremlin API
D) Cassandra API

Answer: C) Gremlin API	*Gremlin API is designed for graph data — perfect for relationships between entities.*

6. Which role is primarily responsible for building data pipelines and transforming raw data?

A) Database Administrator
B) Data Analyst

C) Data Scientist
D) Data Engineer

Answer: D) Data Engineer	*Data Engineers build and maintain data pipelines and infrastructure.*

7. In Power BI, you need your report to always show real-time data without any delay. Which connectivity mode should you use?

A) Import Mode
B) Scheduled Refresh
C) DirectQuery
D) Incremental Refresh

Answer: C) DirectQuery	*DirectQuery queries the source in real-time instead of importing data into Power BI.*

8. Your organisation processes millions of IoT sensor events per second. Which Azure service ingests these events?

A) Azure Data Factory
B) Azure Event Hubs
C) Azure SQL Database
D) Azure Table Storage

Answer: B) Azure Event Hubs	*Event Hubs is designed for high-throughput event ingestion — millions of events per second.*

9. Which file format is best suited for analytical queries on large datasets due to its columnar storage?

A) CSV
B) JSON
C) Avro
D) Parquet

Answer: D) Parquet	*Parquet is columnar — reads only the columns needed, making analytics queries much faster.*

10. Microsoft Fabric stores all data in a single unified storage layer. What is this called?

A) Azure Data Lake
B) OneLake
C) Blob Storage
D) Synapse Pool

Answer: B) OneLake	*OneLake is Fabric's unified storage — one copy of data accessible by all Fabric components.*

QUICK REFERENCE — Service Cheat Sheet

If the question says...	Answer is...
Fully managed SQL, cloud-native app	Azure SQL Database
Migrate on-prem SQL Server with minimal changes	Azure SQL Managed Instance
Need full OS / legacy SQL Server features	SQL Server on Azure VM
NoSQL JSON documents, global distribution	Azure Cosmos DB (NoSQL API)
Existing MongoDB app to Azure	Azure Cosmos DB (MongoDB API)
Graph data, relationships between entities	Azure Cosmos DB (Gremlin API)
Wide-column store, Cassandra workloads	Azure Cosmos DB (Cassandra API)
Simple key-value upgrade from Table Storage	Azure Cosmos DB (Table API)
Store large files, images, videos	Azure Blob Storage
Shared cloud file drive (SMB/NFS)	Azure File Storage
Simple cheap key-value NoSQL	Azure Table Storage
Orchestrate / schedule data pipelines	Azure Data Factory
All-in-one: SQL + Spark + pipelines	Azure Synapse Analytics
Apache Spark, ML, AI, MLflow	Azure Databricks
Real-time SQL stream processing	Azure Stream Analytics
Ingest millions of events per second	Azure Event Hubs
Unified end-to-end analytics (newest)	Microsoft Fabric
Single storage layer in Fabric	OneLake
Business reports and dashboards	Power BI
Raw storage for big data analytics	Azure Data Lake Storage Gen2
Managed Hadoop / Hive / HBase / Kafka	Azure HDInsight
Log analytics, time-series, telemetry	Azure Data Explorer
Real-time data always, no refresh lag	Power BI DirectQuery
Fastest Power BI performance	Power BI Import Mode
Strongest Cosmos DB consistency	Strong consistency
Fastest Cosmos DB consistency	Eventual consistency
Default Cosmos DB consistency	Session consistency
ACID transactions guarantee	Relational / OLTP databases
Columnar file format for analytics	Parquet

If the question says...	Answer is...
Row-based format good for streaming	Avro

EXAM DAY TIPS

- Passing score is **700 / 1000** — you don't need to be perfect.
- Questions are mostly **scenario-based**: 'Which service should you use for X?'
- **Microsoft Fabric** is the newest service — expect 1–3 questions on it.
- Know the difference between **PaaS vs IaaS** for SQL services.
- Know **Cosmos DB APIs** — which API for which use case.
- Know **Batch vs Streaming** and which Azure services handle each.
- Know the 3 **data roles**: DBA, Data Engineer, Data Analyst — and their responsibilities.
- Take the **free official practice assessment** on Microsoft Learn before your exam.
- Use the **exam sandbox** on Microsoft Learn to get familiar with the question interface.

Good luck, Sarvesh! You've got this ■

Practice Assessment → learn.microsoft.com/en-us/credentials/certifications/exams/dp-900